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Hot Tapping of Ethylene Pipelines

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This paper discusses the following problem areas associated with hot tapping pressurized ethylene pipelines: 1) weld cracking, heat affected zone cracking and other common weld defects. 2) blow through, and 3) unstable pipe contents such as ethylene under certain conditions. It discusses the causes or explanation of these problem areas and makes recommendations as to how to avoid or prevent them. This paper references important previous work that has been done on the subject such as the Battelle and the Monsanto studies and combines the conclusions of these studies into one set of recommended procedures. Also contained in the paper is a chart showing actual successful ethylene hot tap job data.

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ABSTRACT

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Also contained in the paper is a chart showing actual successful ethylene hot tap job data.

NOMENCLATURE

HAZ = welding heat affected zone
CE = carbon equivalent percent, %
C = carbon percent, %
Mn = manganese percent, %
Cr = chrome percent, %
Mo = molybdenum percent, %
V = vanadium percent, %
Ni = nickel percent, %
Cu = copper percent, %
P = pressure, psi
S = specified minimum yield strength, psi
t = nominal pipe wall thickness, inch
D = nominal pipe diameter, inch
c = correction factor, inch
V = velocity ft/sec
F = temperature degrees fahrenheit
C_A = concentration of A, parts per million

INTRODUCTION

The procedure for cutting an opening into an operating pipeline (main, storage tank, or other pressure vessel) and connecting branch piping while the line has product under pressure and/or flowing conditions, is referred to as a "hot tap".

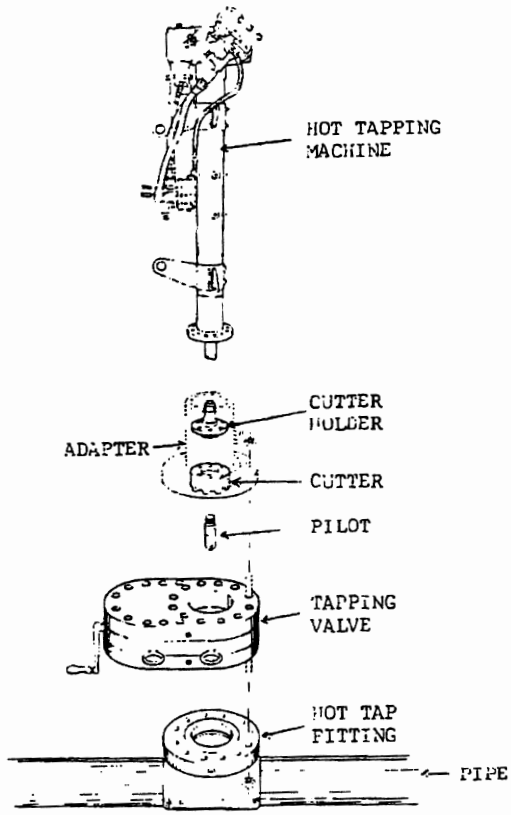

Fig. 1 Typical Hot Tapping Equipment Set-Up

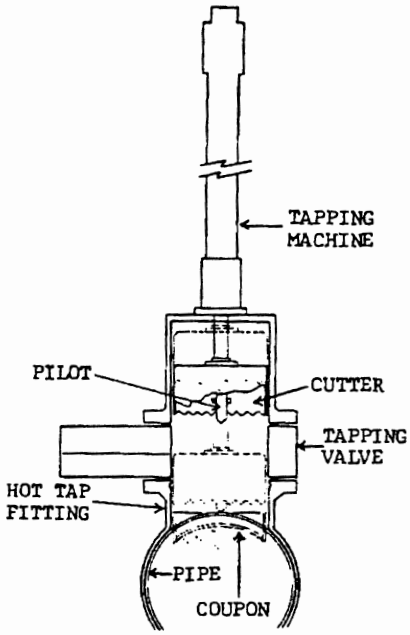

Fig. 2 Hot Tap into Pipe

Welding that is required to attach a branch connection fitting (hot tap fitting) to an operating pipeline is called, "hot tap welding". The vast majority of hot tap welding is manual, metal arc, and is usually completed under less than ideal conditions.

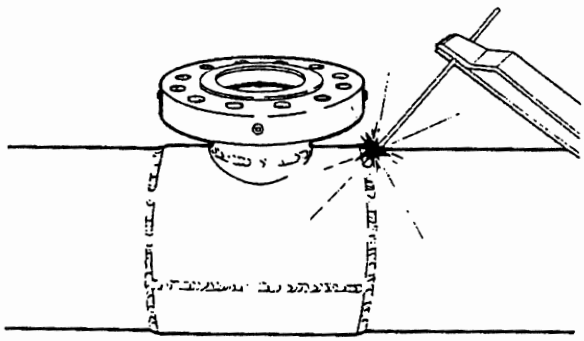

Fig. 3 Hot Tap Welding

The practice of "hot tapping" pressurized pipelines has become widespread throughout the hydrocarbon and chemical processing industry. It has spread from the cross-country pipelines, to the plants and refineries, to such an extent that hundreds of hot tap jobs are performed each day in plants and refineries.

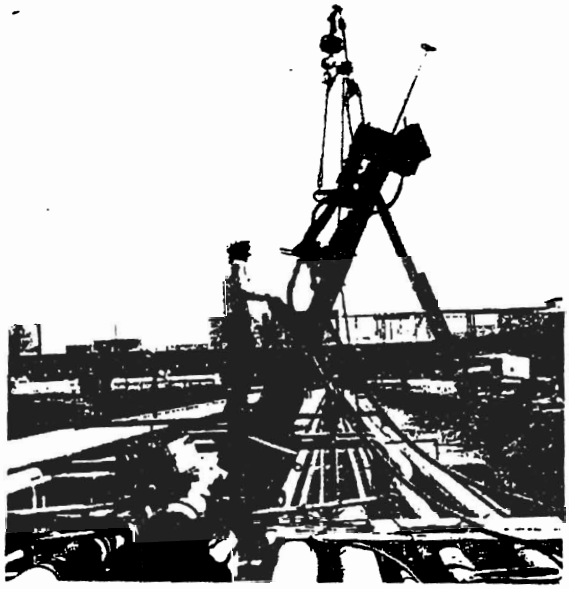

Fig. 4 Large Diameter Hot Tap in Plant

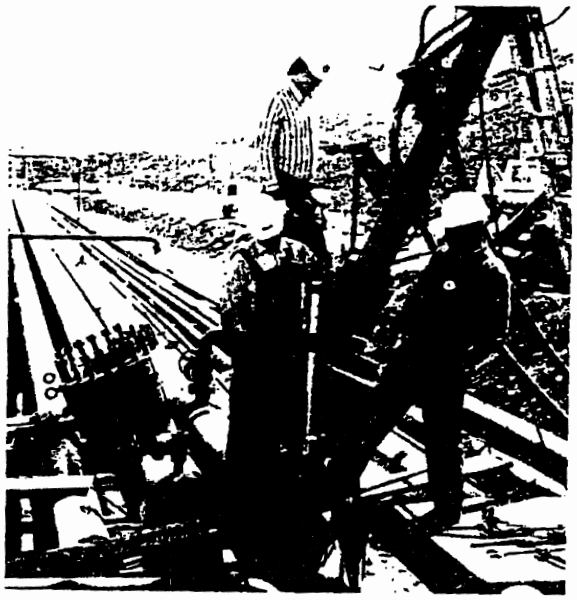

Fig. 5 Small Diameter Hot Tap in Plant

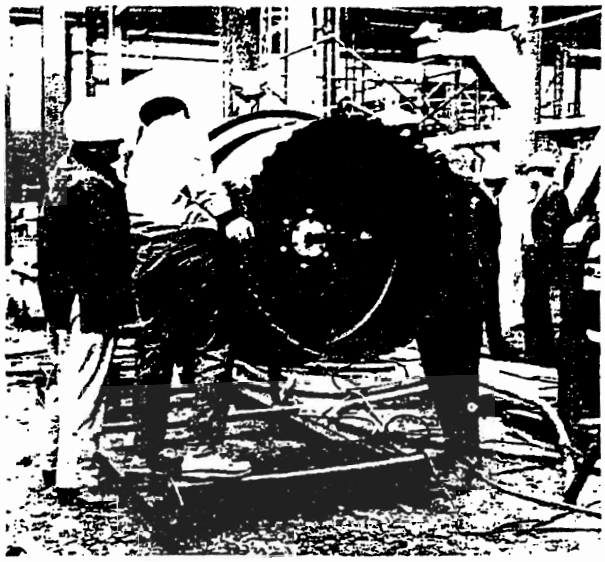

Fig. 6 Installing Cutter on Tapping Machine in Plant

Hot tapping on pipelines containing water, crude oil, natural gas, petroleum products and other common pipelined materials has been done safely for many years. Much hot tapping data has been developed for these materials.

As the practice of hot tapping became more and more accepted in the plant and refinery industry, the question of safety in hot tapping became more complex. Hot tapping in this industry required data and procedures to work on the full range of chemicals and petroleum products.

Several chemicals and products immediately became problems to hot tap either because of the known explosive or decomposition limits or the unknown explosive or decomposition limits. In many cases, test data was available which established safe limits, or indicated hot tap welding or hot tapping should not be done. In many cases, test data was just not available. Ethylene, the subject of this paper was such a product.

Ethylene (C_2H_4), an unsaturated hydrocarbon, is a colorless gas which under certain conditions can be decomposed explosively in the absence of oxygen. It's main use has been in the production of ethylene oxide, chlorinated hydrocarbons, chlorohydrins, esters, and styrene. In addition, it is used as a refrigerant, fruit-ripening agent, and anesthetic (1).

In 1970, based on the lack of limits for hot tapping ethylene, the industry decided to shutdown, ethylene hot tap work until the proper limits could be established.

A group of oil and chemical companies that had an interest in resuming hot tap work on ethylene contracted with Battelle Memorial Institute to study the mechanical welding part of this problem. Monsanto Chemical Company had previously studied the chemical reaction part of the problem and agreed to add this data to the project. This basic data was presented by Monsanto Chemical Company (2) and Battelle Memorial Institute (3) at the 79th National Meeting of AIChE - Houston, Texas in 1975, allowing resumption of hot tapping of ethylene lines.

PROBLEM AREAS

Generally, in almost all hot tapping jobs the main problem areas to be dealt with or avoided, are associated with welding the fitting on the pipeline. If the fitting can be welded to the pipe safely then the hot tapping procedure can be done safely.

There are three main areas that need to be taken into account during the fitting installation stage. They are:

1 Weld cracking, heat affected zone (HAZ) cracking, or other common weld defects.
2 Blow through where the weakened pipe material under the weld and the molten pool is blown out by the internal pipe pressure.
3 Unstable pipe contents.

Common Weld Cracking is caused by the following factors: joint restraint, bead shape, weldability of the material, hydrogen pickup, and high cooling rate.

Heat Affected Zone cracking is usually caused by absorption of atomic hydrogen from the weld metal into the HAZ of the base metal. A rapid cooling rate traps this hydrogen between the crystal lattice and builds up pressure. This pressure combined with the shrinkage and hardening effect of the steels chemistry, causes cracking.

Other common weld defects such as lack of fusion, undercut, and porosity are caused by the correctable welding variables such as improper weld gap, welding speed, low current, electrode size, faulty electrode manipulation, high current, short arc, insufficient puddling time, wrong electrode, etc.

All of the above problems can be prevented by the following:

1 Control the carbon and alloy content of the base metals. The carbon equivalent should be low. A carbon equivalent less than or equal to .43 is suggested on pipe over X-42, one formula for C.E. would be: (4)(5)(6)

$$CE = C + \frac{Mn}{6} + \frac{Cr + Mo + V}{5} + \frac{Ni + Cu}{15} \qquad (1)$$

2 Employ a low hydrogen process.
3 Control the heat input.
4 Use the proper sequence of welds (on full encirclement fittings longitudinal welds first, then one end circumferential weld, then the other). (See Figure 7)
5 Qualify welders and weld procedures to applicable codes and make sure procedures are followed.
6 Heat area on the pipe where fitting is to be installed to remove moisture from the line.
7 Inspection and testing of the field weld is an important procedure after the welding is complete. It is a good idea to pressure test the hot tap fitting before the hot tap and to nondestructively examine the welding.

Blow through is a term describing a condition where the weakened pipe material under the weld, and

the molten pool, is blown out by the internal pipe pressure. On non-full encirclement fittings, tests have shown that blow through would most likely occur on the weld running longitudinal to the pipeline (3). On a full encirclement fitting the only welds made directly to the pipeline are the end circumferential weld.

From previous tests, it has been shown that the possible causes of blow through could be listed as follows (3) (7):

1 Type of electrode (deeper penetration vs shallow penetration electrode).
2 Electrode manipulation (electrode being left in one spot too long).
3 Direction of welding in relation to hoop stress on the pipeline. Welding perpendicular to pipe hoop stress has been found to be more likely to produce a blow through condition.
4 Pipe wall thickness - tests have shown blow through possible with .156 inch wall pipe.
5 Pipe internal pressure or stress level in pipe. Tests have shown blow through possible on pipe wall .188 inch with a 72% stress level.
6 High heat input.

Work done thus far, in several important areas of study on preventing blow through, allow some guidelines to be drawn. Additional technical work needs to be done on this important subject.

Present guidelines to prevent blow through are as follows:

1 A shallow penetration electrode (low hydrogen) should be used.
2 The contents of the line should be flowing.
3 The electrode should be manipulated so that there is a heat balance between the fitting and the pipe.
4 The weld attaching the fitting to the pipe should be circumferential to the pipe. It should not be perpendicular to the hoop stress of the pipeline.
5 The minimum pipe wall should be .156 inch or greater. If the weld must be longitudinal to the pipe then the minimum wall should be .188 inch with less than a 72% stress level.
6 The pressure in the pipeline should be reduced according to the ASME Gas Piping Standards Committee's recommended formula for welding split repair sleeves on gas pipelines (8). (Other industry or company pressure reduction formulas could be used.) ASME formula:

$$P = \frac{2S\,(t-c)\,0.72}{D} \qquad (2)$$

S = specified minimum yield strength, psi
t = nominal pipe wall thickness, inch
D = nominal pipe diameter, inch
c = correction factor, inch - for which present committee thinking reflects a value of 3/32 inch (allowance for heated metal loss of strength).

The ASME formula is thought to be conservative.

Factors to be considered concerning unstable pipe contents are normally related to pipe contents that become unstable with the addition of heat and/or non-flowing conditions. Some examples of this type of problem are as follows:

Combustible mixtures such as air and fuel or oxygen and fuel of the mixture percentages that could allow ignition and propagation of oxidation (sustained combustion) reaction with the addition of heat, should not be welded upon or hot tapped.

Oxygen, oxygen enriched air or compressed air could allow ignition and propagation of oxidation with the addition of heat and fuel. These should not be welded upon if there is any possibility of fuel being present.

Chlorine can present similar problems.

Hydrogen, hydrogen mixtures, and caustic present a different type problem. The main problem with welding on these materials is the possible cracking that can occur in the weld metal or heat affected zone. Welding and hot tapping on lines containing these materials should be avoided if there is any possibility that the hydrogen could get into the molten weld metal, or the caustic could cause stress corrosion cracking in the pipe wall.

No flow conditions can allow the contents of a pipe to reach a higher temperature than with flow conditions. Pipe contents with impurities (oxygen) at higher temperatures are more likely to ignite. Flow conditions help to dissipate heat but serve to speed up the quench rate which is a factor in weld cracking problems.

In gases, no flow conditions are potentially hazardous. Conversely, high flow rates have rarely been found to present problems. A minimum velocity of 1.3 ft/sec on a gas line should be maintained to carry the welding heat away. A maximum flow rate does not appear critical within the limits of practicability (9).

In liquids, a no flow rate condition can also be potentially hazardous. Unlike gases, high flow rates in liquids do present problems.

1 More heat input is required to ensure full weld fusions.
2 High flow rates tend to result in too rapid quenching which contributes to cracking problems.

To avoid these two problems it is recommended that a minimum velocity of 1.3 ft/sec and a maximum of 4 ft/sec be maintained on liquid lines (9).

It should be noted here that preheat and post weld treatment can sometimes be used to allow welding outside these recommendations.

An unstable pipe content that has been the subject of much study the last few years is ethylene. From the studies made by Battelle Memorial Institute (2) and Monsanto (3) in 1975 the following material limits were derived:

1 Ethylene pipelines can be hot tapped up to a maximum of 1200 psi pressure at ethylene temperatures below 300 F, with no possibility of self-sustained ethylene decomposition.

2 Full encirclement hot tap fittings can be welded by competent skilled welders to gas pipelines with a minimum wall thickness of .156 inch, operating at a maximum pressure equivalent to 72% of specified minimum yield strength of the pipe material, without blow-out or burn through (X52 - 20 inch - .156 inch wall - maximum pressure = 584 psi) using both E 6010 and E 7018 welding electrodes.
3 A saddle reinforced fitting, or any fitting where the fillet weld runs longitudinal on the pipe, cannot be welded safely on pipe with a wall thickness less than .188 inch, with the internal pressure equivalent to 72% of specified minimum yield strength of the pipe material. This applies for both E 6010 and E 7018 welding electrodes.
4 E 7018 (low hydrogen) welding electrodes are safer to use then E 6010 electrodes, because they do not penetrate as deep and produce lower through wall temperatures.
5 Ethylene with a concentration of 1000 ppm oxygen is as safe to hot tap and stopple as 5 ppm oxygen.
6 All of the five statements above apply to static pipelines; flowing pipelines should improve safety factors.

Good engineering practice calls for safety factors, or a safety pad, to be included in any operating limits which will be used in a variety of field applications. Toward this end, the following recommended limits were established by T. D. Williamson, Inc.

1 All ethylene jobs require a full encirclement fitting with a welding back-up strip behind the longitudinal welds.

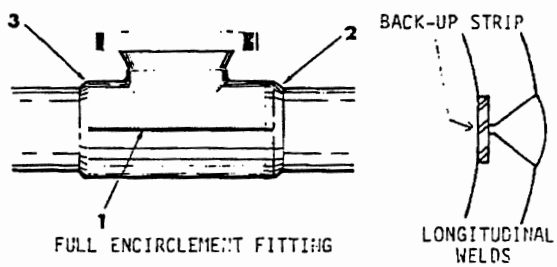

Fig. 7

2 Pipeline maximum or minimum operating limits.
A Maximum temperature - 250 degree F
B Maximum pressure (whichever is lower).
(1) 1100 psi
(2) Equivalent of 72% of specified minimum yield strength of pipe material.
(3) Applicable code restriction, for example:
(a) ASME Gas Piping Standards Committee recommended formula for allowable pressure during repair.

$$P = \frac{2S\,(t-c)\,0.72}{D} \qquad (2)$$

P = internal pressure, psi
S = specified minimum yield strength, psi
t = nominal pipe wall thickness, inch
D = nominal pipe diameter, inch
c = correction factor, inch, for which present committee thinking reflects a value of 3/32.

(b) British Pipeline Industry Guild's Code requirement of allowable pressure during repair of pipelines.
(c) Other industry or company pressure reduction formulas.

C The normal working pressures of hot tapping equipment.
D Minimum wall thickness of pipe - .188 inch
E Minimum yield strength of pipe - 30,000 psi or greater.

3 Low hydrogen electrodes should be required to weld the fitting or the pipe because of shallow penetration and lower through wall temperature characteristics.
4 Some flow in pipeline during welding and hot tap stage is a good practice.
5 Ethylene purity - the maximum allowable concentration of oxygen is 1000 ppm, due to the limits of testing.

Using the above limits T. D. Williamson, Inc. has completed many ethylene hot taps successfully and safely. The following chart summarizes a few of those jobs.

TYPE OF WORK	DATE	SIZE	PRESSURE	TEMPERATURE °F
Hot tap and plug	1966	12"	500 psi	Ambient
Hot tap and plug	1967	3"	60 psi	Ambient
Hot tap and plug	1967	4"	600 psi	Ambient
Hot tap and plug	1967	4"	300 psi	Ambient
Hot tap and plug	1967	3"	60 psi	Ambient
Hot tap and plug	1968	4"	400 psi	Ambient
Hot tap and plug	1968	3"	60 psi	Ambient
Hot tap	1968	4"	600 psi	Ambient
Hot tap and plug	1968	3"	100 psi	Ambient
Hot tap	1969	4"	1100 psi	Ambient
Hot tap	1973	4"	450 psi	Ambient
Hot tap and plug	1973	6"	450 psi	Ambient
Hot tap and plug	1974	4"	300 psi	Ambient
Hot tap and plug	1974	8"	700 psi	Ambient
Hot tap and plug	1974	4"	300 psi	Ambient
Hot tap	1974	8"	1000 psi	Ambient
Hot tap and plug	1976	6"	300 psi	Ambient
Hot tap and plug	1976	6"	200 psi	Ambient

Fig. 8 Ethylene Hot Tap Job Data

CONCLUSION

In conclusion, present day technology relating to hot tapping of ethylene pipelines, which has been developed over the past few years, is being applied by the industry as a useful procedure, saving time and money. The operating limits established have proved to be safe when applied with good engineering judgement. Any future work on ethylene decomposition limits or the subject of hot tap welding would be welcomed by the industry to further broaden the base that has already been established.

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CTS Notes :

1. Operating velocity indicated above is higher than permissible for hot tappings. FCC operation is to ensure operating velocity between 0.4-1.7 m/sec max. during welding as per hot tapping DEP guidelines.
For carrying out hot tapping at velocity > 1.7 m/sec, heat loss tests should be performed at the actual hot tapping location. In the event of rapid cooling, special interpass heating techniques may be needed or the welding procedure amended.

2. Actual analysis data is not available. VMP design data is provided from UOP Sch A

3. The fluid contains hazardous/unstable substances (H2S and ethylene/propylene). However, hot tapping may be carried out with precautions and following guidelines as mentioned in attached tech. paper from M/s TDW.
Refer guidelines on Pg 4, 5 and 6 of the technical paper (Highlighted).

FCC operations to carry out risk analysis and ensure safety during the hot tapping work.